

Data Upload

Upload your business dataset.

Supported formats:

-  ZIP
-  CSV
-  Excel (.xlsx)

Choose Dataset

 Online...Orders.xlsx 105.2KB

+

 Excel Uploaded Successfully

Shape: (1200, 14)

Columns:

```
▼ [
  0 : "OrderID"
  1 : "Date"
  2 : "CustomerID"
  3 : "Product"
  4 : "Quantity"
  5 : "UnitPrice"
  6 : "ShippingAddress"
  7 : "PaymentMethod"
  8 : "OrderStatus"
  9 : "TrackingNumber"
 10 : "ItemsInCart"
 11 : "CouponCode"
 12 : "ReferralSource"
 13 : "TotalPrice"
]
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	Paym
0	ORD200000	2023-01-04 00:00:00	C72649	Monitor	5	570.62	928 Main St	Debit
1	ORD200001	2024-08-23 00:00:00	C75739	Phone	2	151.35	823 Main St	Onlin
2	ORD200002	2024-02-27 00:00:00	C81728	Tablet	5	550.68	512 Main St	Credi
3	ORD200003	2023-10-15 00:00:00	C33540	Chair	1	273.19	275 Main St	Debit
4	ORD200004	2025-05-08 00:00:00	C81840	Printer	4	626.01	668 Main St	Onlin

30

Dashboard

Data Profile

Dataset Understanding

```
{
  "type": "table" |>
  "data_columns": [
    0: "date"
  ]
  "numeric_columns": [
    0: "quantity"
    1: "unitPrice"
    2: "discount"
    3: "totalPrice"
  ]
  "categorical_columns": [
    0: "orderId"
    1: "customerId"
    2: "product"
    3: "shippingAddress"
    4: "paymentMethod"
    5: "orderStatus"
    6: "trackingNumber"
    7: "couponCode"
    8: "referralSource"
  ]
}
```

Semantic Model

```
{
  "data": "table"
  "customer": "customerId"
  "product": "product"
  "quantity": "quantity"
  "unitPrice": "unitPrice"
  "discount": "discount"
  "totalPrice": "totalPrice"
  "paymentMethod": "paymentMethod"
  "transaction": "paymentMethod"
  "status": "orderStatus"
}
```

Generated Metrics

```
{
  "Total Records": 1200
  "Total Quantity": 1200
  "Total Orders": 1200
  "Customers": 1200
}
```

Dataset Intelligence

```
{
  "rows": 1200
  "columns": [
    "numeric_columns": [
      0: "quantity"
      1: "unitPrice"
      2: "discount"
      3: "totalPrice"
    ]
    "categorical_columns": [
      0: "orderId"
      1: "customerId"
      2: "product"
      3: "shippingAddress"
      4: "paymentMethod"
      5: "orderStatus"
      6: "trackingNumber"
      7: "couponCode"
      8: "referralSource"
    ]
  ]
  "data_columns": [
    0: "date"
  ]
  "binary_columns": [
  ]
  "id_columns": [
    0: "orderId"
    1: "customerId"
  ]
  "measure_columns": [
    0: "quantity"
    1: "unitPrice"
  ]
  "dimension_columns": [
    0: "product"
    1: "paymentMethod"
    2: "orderStatus"
    3: "couponCode"
    4: "referralSource"
  ]
}
```

Dataset loaded successfully

Preview

orderId	date	customerId	product	quantity	unitPrice	shippingAddress	payment
0	2023-01-01 00:00:00	CT0493	Monitor	5	578.62	828 Main St	Debit
1	2023-08-01 23:00:00:00	CT0789	Phone	2	155.89	421 Main St	Online
2	2024-02-27 00:00:00	CA1726	Tablet	5	350.08	111 Main St	Credit
3	2023-01-13 00:00:00	C33840	Chair	3	375.12	175 Main St	Debit
4	2023-05-05 00:00:00	CA1840	Printer	4	426.81	666 Main St	Online

Dataset Profile

Rows

1200

Columns

14

Missing Values

309

Duplicates

0

Warning (8%)

0.21

Numeric Columns

4

Categorical Columns

9

Date Columns

1

Column Information

Columns	Data Type	Missing Values	Unique Values
orderId	orderId	0	1200
date	Date	0	015
customerId	customerId	0	1200
product	product	0	7
quantity	quantity	0	5
unitPrice	unitPrice	0	1889
shippingAddress	shippingAddress	0	655
paymentMethod	paymentMethod	0	5
orderStatus	orderStatus	0	5
trackingNumber	trackingNumber	0	1200

Dataset Summary

orderId	date	customerId	product	quantity	unitPrice	shippingAddress	payment
value1	1200	1200	1200	1200	1200	1200	1200
unique1	1200	None	1200	7	None	None	655
top	2023-08-01	None	CT0879	Printer	None	None	513 Main St
freq	2	None	2	185	None	None	0
mean	None	2014-02-23 16:00:00	None	None	230450	206.4138	None
std	None	2013-08-01 00:00:00	None	None	1	11.39	None
25%	None	2013-08-01 00:00:00	None	None	2	106.0625	None
50%	None	2014-02-23 00:00:00	None	None	3	384.11	None
75%	None	2014-01-02 00:00:00	None	None	4	811.07	None
max	None	2013-08-30 00:00:00	None	None	3	893.03	None

Missing Values

0

Coupon Code

200

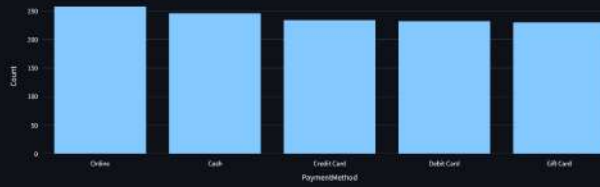
Quantity Trend



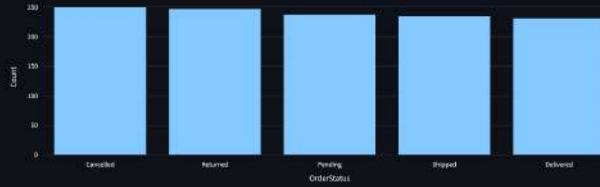
Product Distribution



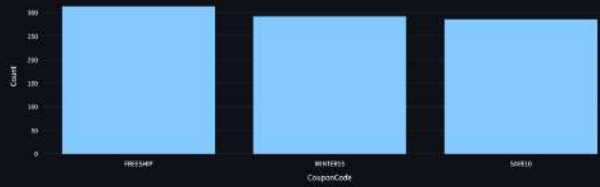
PaymentMethod Distribution



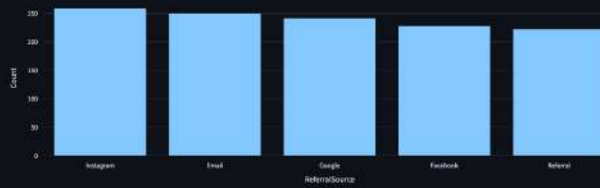
OrderStatus Distribution



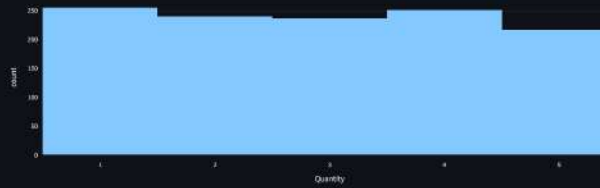
CouponCode Distribution



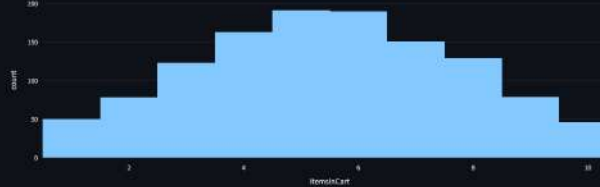
ReferralSource Distribution



Quantity Distribution



ItemsInCart Distribution



🗯️ Ask Your Data

Ask a question about your data

how to increase sales

To increase sales, a professional data analyst would recommend a multi-faceted approach, leveraging your existing sales data to identify opportunities and optimize strategies. Here's how you can use your dataset to gain actionable insights:

Key Areas to Analyze and Actions to Take: ↔️

1. Product Performance Analysis:

- **What to analyze:**
 - **Top-selling products:** Calculate the total `Quantity` sold and `TotalPrice` generated for each `Product`.
 - **Slow-moving products:** Identify products with low sales volume or revenue.
 - **Profitability per product:** While `UnitPrice` is available, knowing `CostPrice` would allow for actual profit calculation. Without it, `TotalPrice` per product is the best proxy for revenue generated.
- **Business Insight & Action:**
 - **Promote best-sellers:** Invest more in marketing and stock for your most popular items.
 - **Bundle or discount slow-movers:** Pair them with best-sellers or offer strategic discounts to clear inventory and attract buyers.
 - **Optimize product offerings:** Consider discontinuing consistently underperforming products or re-evaluating their pricing/marketing.

2. Customer Segmentation and Behavior:

- **What to analyze:**
 - **High-value customers:** Identify customers (`CustomerID`) with the highest cumulative `TotalPrice`.
 - **Purchase frequency:** Track how often specific `CustomerIDs` make purchases.
 - **Items in cart vs. purchased:** Analyze the average `ItemsInCart` and compare it to actual `Quantity` purchased to understand potential cart abandonment issues or opportunities for upselling.
- **Business Insight & Action:**
 - **Loyalty programs:** Reward and retain your high-value customers with exclusive offers.
 - **Personalized recommendations:** Use their purchase history to suggest related products or remind them of items in their `ItemsInCart`.
 - **Re-engagement campaigns:** Target customers who haven't purchased recently.

3. Promotional and Marketing Effectiveness:

- **What to analyze:**
 - **Coupon code effectiveness:** Measure the `TotalPrice` and `quantity` sold associated with each `CouponCode`.
 - **Referral source performance:** Determine which `ReferralSource` (e.g., Instagram, Email, Facebook) generates the most sales (`TotalPrice` and `Quantity`).
- **Business Insight & Action:**
 - **Optimize coupon strategy:** Replicate successful `CouponCode` campaigns and refine or discontinue underperforming ones.
 - **Targeted marketing investment:** Allocate more marketing budget to the `ReferralSources` that demonstrate the highest return on investment.

4. Order Status and Fulfillment Optimization:

- **What to analyze:**
 - **Cancellation and return rates:** Calculate the percentage of orders with `OrderStatus` as 'Cancelled' or 'Returned' by `Product`, `CustomerID`, or over `Date` periods.
 - **Delivery times:** While not directly in your columns, 'Shipped' to 'Delivered' could imply time, if you had a 'DeliveryDate' column.
- **Business Insight & Action:**
 - **Identify pain points:** High cancellation/return rates for specific products might indicate quality issues, misleading descriptions, or problems with the ordering process. Investigate reasons for 'Cancelled' orders (e.g., high shipping costs, long delivery times).
 - **Improve customer satisfaction:** Streamline your shipping and delivery process to ensure timely and accurate fulfillment, reducing returns and increasing customer trust.

5. Seasonal and Trend Analysis:

- **What to analyze:**
 - **Sales trends over time:** Group `TotalPrice` and `Quantity` by `Date` (e.g., month, quarter, year) to identify peak sales periods and seasonality.
- **Business Insight & Action:**
 - **Strategic planning:** Forecast demand, plan inventory, and schedule marketing campaigns around identified peak sales seasons.
 - **Identify growth opportunities:** Understand if sales are growing, stagnant, or declining over time and investigate underlying reasons.

By systematically analyzing these aspects of your data, you can uncover specific areas for improvement and implement data-driven strategies to effectively increase your sales.

AI Insights

Gemini Business Insights

As a Senior Business Intelligence Analyst, I've reviewed the provided dataset snapshot and KPIs to offer a business analysis.

Dataset Overview: The dataset contains 1200 records with 14 columns, providing detailed information about individual orders. Key data points include order specifics (`OrderID` , `Date` , `Product` , `Quantity` , `UnitPrice` , `TotalPrice`), customer information (`CustomerID`), fulfillment details (`ShippingAddress` , `OrderStatus` , `TrackingNumber`), payment methods (`PaymentMethod`), and marketing/sales drivers (`ItemsInCart` , `CouponCode` , `ReferralSource`).

KPIs:

- Total Records: 1200
- Missing Values: 309

1. Key Business Insights

- Comprehensive Order Tracking:** The dataset allows for a detailed understanding of individual orders from placement (`Date`) through fulfillment (`OrderStatus` , `TrackingNumber`) to financial transaction (`TotalPrice` , `PaymentMethod`).
- Customer Behavior Indicators:** `CustomerID` enables analysis of repeat purchases (if aggregated), while `ItemsInCart` provides insight into initial customer intent versus final purchase (`Quantity` of specific product).
- Marketing Channel Visibility:** The `ReferralSource` column offers direct insights into which channels drive customer acquisition, facilitating marketing strategy evaluation.
- Promotional Effectiveness:** `CouponCode` provides a mechanism to track the usage and potential impact of promotional campaigns.
- Revenue Tracking:** `TotalPrice` and the combination of `Quantity` and `UnitPrice` allow for direct measurement of revenue generated per order and per product.
- Order Fulfillment Lifecycle:** The `OrderStatus` column (e.g., Shipped, Cancelled, Returned, Delivered) offers a clear view of the order journey and potential areas for operational improvement.

2. Important Trends

- Time-Based Performance:** With the `Date` column available, the business can analyze trends in order volume, total revenue, average order value, popular products, and customer acquisition (by `ReferralSource`) over time (e.g., monthly, quarterly, annually).
- Customer Acquisition Trends:** By analyzing `ReferralSource` over time, it's possible to identify which marketing channels are gaining or losing effectiveness in attracting customers.
- Order Status Trends:** Trends in `Cancelled` or `Returned` orders can be monitored over time to identify periods of increased issues or improved performance in product quality or fulfillment processes.
- Payment Method Preferences:** The `PaymentMethod` column allows for tracking changes in how customers prefer to pay, which can inform payment system optimizations.
- Coupon Usage Trends:** Analysis of `CouponCode` usage over time can reveal the impact and popularity of different promotions.

3. Risks or Problems

- Significant Data Quality Issue (Missing Values):** The most prominent risk is the 309 missing values out of 1200 records. This represents approximately 25.75% of the dataset. Such a high proportion of missing data can severely compromise the accuracy and reliability of any analysis, leading to skewed insights or an inability to draw firm conclusions. Without knowing which specific columns are affected, it's difficult to pinpoint the exact impact, but it suggests a systemic data collection or integration issue.
- Incomplete Customer View:** While `CustomerID` is present, there's no additional demographic or behavioral data (e.g., customer tenure, segmentation) that would enable deeper customer lifetime value (CLV) analysis or personalized marketing efforts.
- Limited Financial Scope:** Although `TotalPrice` is available for revenue tracking, the absence of cost data (e.g., cost of goods sold, shipping costs, marketing spend per channel) prevents profitability analysis or precise ROI calculations.
- Ambiguity of `ItemsInCart` vs. `Quantity`:** While these columns offer potential insights, without further data or context, it's unclear if `ItemsInCart` consistently reflects the total distinct items or total units in the cart, and how this relates to final purchased `Quantity` .
 - Possible Interpretation:* If `ItemsInCart` is often higher than the `Quantity` of the listed `Product` , it could indicate abandoned items in the cart or cross-selling opportunities that were not realized in the final purchase of the listed product. However, this requires further detailed analysis.

4. Business Recommendations

- Prioritize Data Quality Enhancement:**
 - Immediate Action:** Investigate the source and nature of the 309 missing values. Determine which columns are most affected and the reasons for incompleteness.
 - Long-term Strategy:** Implement robust data validation rules and improve data collection processes at the source to prevent future data gaps. Data completeness is critical for reliable BI.
- Leverage Data for Comprehensive Trend Analysis:**
 - Conduct in-depth time-series analysis on sales volume, total revenue, average order value, and product performance using the `Date` column.
 - Analyze trends in `ReferralSource` effectiveness to optimize marketing budget allocation and strategies over time.
 - Monitor `OrderStatus` trends (e.g., cancellation rates, return rates) on a rolling basis to identify operational bottlenecks or product quality issues quickly.
- Deep Dive into Order Statuses:**
 - Analyze `Cancelled` and `Returned` orders to identify common characteristics (e.g., specific products, customer segments, payment methods, `ReferralSource`) and potential root causes. This can lead to improvements in product descriptions, quality control, or fulfillment processes.
- Evaluate Marketing & Promotional Effectiveness:**
 - Analyze `ReferralSource` data to pinpoint the most effective customer acquisition channels and reallocate marketing resources accordingly.
 - Evaluate `CouponCode` usage and its correlation with `Quantity` and `TotalPrice` to understand the actual impact and ROI of promotional campaigns (assuming `TotalPrice` reflects the discounted value).
- Explore Customer Behavior and Cart Dynamics:**
 - Analyze the relationship between `ItemsInCart` , `Quantity` , and `OrderStatus` .
 - Possible Interpretation:* If `ItemsInCart` is significantly higher than `Quantity` in cancelled orders, it might indicate issues during the checkout process or last-minute changes in customer intent, warranting a review of the checkout flow.
- Data Enrichment for Advanced Analytics:**
 - Consider integrating additional data points such as product cost (for profitability analysis), customer demographics, and customer feedback (e.g., satisfaction scores) to enable more holistic financial, customer experience, and segmentation analyses.



Business Forecasting



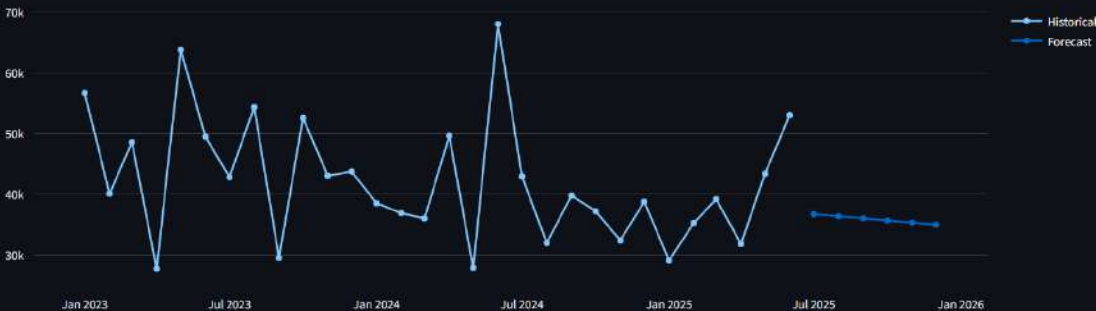
Forecast Capability

Forecast Available

Detected Dataset: GENERIC



TotalPrice Forecast



Forecast Summary

Current Value

53047.4

Future Prediction

35001.13

Expected Change

-34.02%

Forecast trend: Decreasing



AI Forecast Explanation

Business Forecast Explanation: ForecastValue

- Forecast Observation** The 'ForecastValue' metric is projected to experience a significant decline, decreasing from its historical last value of 53047.4 to a future predicted value of 35001.13. This represents an expected reduction of -34.02%.
- Trend Explanation** The overall trend indicated by this forecast for 'ForecastValue' is definitively decreasing, reflecting the substantial predicted drop from the historical baseline.
- Assumptions and Limitations**
 - Assumptions:** This forecast assumes the accuracy and reliability of the underlying predictive model and the historical data used to generate it.
 - Limitations:** The provided information does not include the specific reasons or contextual drivers behind this predicted decrease. Consequently, this explanation cannot account for potential impacts from external market conditions, internal operational changes, or strategic initiatives that may influence the 'ForecastValue'. The analysis is based solely on the quantitative forecast data provided.



Future Prediction

	TimeIndex	ForecastValue	Date
0	30	36746.89	2025-07-01 00:00:00
1	31	36397.7389	2025-08-01 00:00:00
2	32	36048.5878	2025-09-01 00:00:00
3	33	35699.4367	2025-10-01 00:00:00
4	34	35350.2856	2025-11-01 00:00:00
5	35	35001.1345	2025-12-01 00:00:00

Executive Business Report

Selected Dataset Type: GITHUB

Business Overview

Total Records	Total Value	Average Value	Total Quantity
1200	1264761.96	1053.97	3535
Average Price	Missing Values		
356.41	309		

Executive Summary

Executive Business Intelligence Report

Date: October 26, 2023 To: Business Leadership From: Senior Business Intelligence Analyst Subject: Overview of Operational Performance and Strategic Insights

This report provides a high-level overview of our operational performance based on the available dataset, identifying key metrics, potential trends, and areas for strategic focus.

1. Overall Business Performance

Our operations involve the processing of discrete transactions, each tied to a unique order and customer interaction.

- Transaction Volume:** A total of 1,200 orders/transactions have been recorded and processed.
- Total Processed Value:** The cumulative monetary value associated with these transactions amounts to \$1,264,761.96.
- Average Transaction Value:** On average, each transaction holds a value of \$1,053.97. This metric indicates the typical scale of individual customer engagements.
- Item Quantity:** A total of 3,535 items were processed across all transactions, reflecting the volume of goods or services handled.
- Average Unit Price:** The average unit price of items involved in these transactions is \$356.41.

These figures represent the overall scale and financial impact of our operational activities.

2. Key Trends and Patterns

While detailed trend analysis requires further data aggregation, the available dataset columns suggest several areas for potential insight:

- Temporal Analysis Potential:** The presence of a 'Date' column allows for comprehensive time-series analysis. This can be utilized to understand fluctuations in transaction volume, processed value, item quantity, and product performance over time (e.g., daily, weekly, monthly, yearly trends). Specific temporal trends are not pre-calculated in the provided KPIs and therefore cannot be reported directly at this time.
- Customer Engagement & Sourcing:** The 'CustomerId' column allows for tracking individual customer activity. 'ReferralSource' and 'CouponCode' provide insights into acquisition channels and promotional effectiveness. 'ItemsInCart' offers a view into customer shopping behavior prior to finalisation.
- Operational Flow:** The 'OrderStatus' column (e.g., Shipped, Delivered, Cancelled, Returned, as seen in sample data) is critical for understanding the efficiency and success rate of our order fulfillment process.
- Product Insights:** The 'Product' column, combined with 'Quantity' and 'UnitPrice', enables analysis of item popularity, value contribution, and inventory management considerations.
- Payment & Logistics:** 'PaymentMethod' and 'ShippingAddress' can provide insights into customer preferences and logistical patterns.

Possible Interpretation: The variations in 'OrderStatus' observed in sample data (e.g., 'Cancelled', 'Returned') indicate that not all initiated transactions are completed successfully or to customer satisfaction, pointing to potential areas for process improvement.

3. Major Risks or Concerns

- Data Quality - Missing Values:** A significant concern is the presence of 309 missing values across the dataset. This represents approximately 25.75% of the total records (309 out of 1200).
 - Impact:** Missing data can lead to incomplete analysis, inaccurate reporting, and potentially flawed decision-making. Depending on which columns contain these missing values, it could hinder customer profiling, product analysis, or operational tracking.
- Operational Inefficiencies (Potential):** As noted in the 'Key Trends' section, the existence of 'Cancelled' and 'Returned' orders in the sample data suggests potential for improvement in processes related to order accuracy, product quality, or delivery. Without aggregated figures, the scale of this issue cannot be quantified, but it warrants investigation.
- Lack of Granular Performance Metrics:** While aggregate metrics are provided, the absence of pre-calculated, dimension-specific performance indicators (e.g., total value by product, cancellation rate by payment method, customer lifetime value) limits deeper immediate insights.

4. Strategic Recommendations

Based on the current analysis, the following strategic recommendations are proposed:

- Prioritize Data Quality Enhancement:**
 - Action:** Immediately investigate the source and nature of the 309 missing values. Identify which columns are affected and implement data validation rules or processes to minimize future occurrences.
 - Benefit:** Improved data integrity will lead to more reliable insights and better-informed strategic decisions.
- Conduct In-Depth Operational Performance Analysis:**
 - Action:** Leverage the 'Date' and 'OrderStatus' columns to perform time-series analysis of order statuses (e.g., tracking cancellation rates, return rates, and delivery success over time). Analyze 'PaymentMethod' and 'ReferralSource' alongside 'OrderStatus' to identify correlations.
 - Benefit:** Identify bottlenecks, common reasons for cancellations/returns, and areas for process optimization to enhance operational efficiency and customer satisfaction.
- Perform Comprehensive Product & Customer Behavior Analysis:**
 - Action:** Analyze 'Product' performance (e.g., most purchased, highest total value) and correlate with 'CustomerId', 'ItemsInCart', 'CouponCode', and 'ReferralSource'.
 - Benefit:** Inform product strategy, marketing campaigns, and customer retention efforts by understanding what customers buy, how they discover us, and their purchasing habits.
- Develop a Centralized BI Dashboard:**
 - Action:** Create an interactive dashboard that visualizes key metrics (Total Value, Total Quantity, Average Value) and allows for drill-down analysis by 'Date', 'Product', 'CustomerId', 'OrderStatus', 'PaymentMethod', and 'ReferralSource'.
 - Benefit:** Provide ongoing, real-time insights to leadership and operational teams, fostering a data-driven culture and enabling proactive decision-making.

This report provides a foundational understanding of our current operational landscape. Implementing these recommendations will enable a more granular and actionable strategic approach.

SEMANTIC CHECK

```
{
  "data": "Data"
  "customer": "CustomerID"
  "product": "Product"
  "quantity": "Quantity"
  "price": "UnitPrice"
  "revenue": "TotalPrice"
  "referral": "ReferralSource"
  "status": "OrderStatus"
  "transaction": "PaymentMethod"
  "payment_method": "PaymentMethod"
  "order_status": "OrderStatus"
  "forecast_target": "TotalPrice"
}
```

Forecast: TotalPrice



Future Prediction

Date	ForecastValue
2023-07-01 00:00:00	34746.89
2023-08-01 00:00:00	36297.7589
2023-09-01 00:00:00	36048.5618
2023-10-01 00:00:00	35899.4367
2023-11-01 00:00:00	35585.2836
2023-12-01 00:00:00	35082.1345